

Hurdle and zero-inflated models - Handout

1 Exercise 1

A random variable Y is said to follow a negative binomial distribution, denoted as $\mathcal{NB}(\mu, \vartheta)$, when its probability mass function (p.m.f.) is

$$\mathbb{P}(Y = y; \vartheta, \mu) = \frac{\Gamma(y + \vartheta)}{\Gamma(\vartheta)\Gamma(y + 1)} \left(\frac{\vartheta}{\mu + \vartheta} \right)^\vartheta \left(\frac{\mu}{\mu + \vartheta} \right)^y,$$

where $\Gamma(u) = \int_0^\infty t^{u-1} \exp(-t) dt$ denotes the gamma function, $\mu > 0$, $\vartheta > 0$ and $y = 0, 1, 2, \dots$

- a) Give the p.m.f. of a random variable distributed as a truncated $\mathcal{NB}(\mu, \vartheta)$, *i.e.* with strictly positive outcomes.

We are interested in finding the expression of :

$$\mathbb{P}(Y = y | Y > 0; \mu, \vartheta) = \frac{\mathbb{P}(Y = y, Y > 0; \mu, \vartheta)}{\mathbb{P}(Y > 0; \mu, \vartheta)} = \frac{\mathbb{P}(Y = y, Y > 0; \mu, \vartheta)}{1 - \mathbb{P}(Y = 0; \mu, \vartheta)}$$

It can be shown that:

$$\mathbb{P}(Y = 0; \mu, \vartheta) = \left(\frac{\vartheta}{\mu + \vartheta} \right)^\vartheta$$

Yielding :

$$\mathbb{P}(Y = y | Y > 0; \mu, \vartheta) = \frac{\Gamma(y + \vartheta)}{\Gamma(\vartheta)\Gamma(y + 1)} \frac{\vartheta^\vartheta}{(\mu + \vartheta)^\vartheta - \vartheta^\vartheta} \left(\frac{\mu}{\mu + \vartheta} \right)^y$$

- b) Define a hurdle model for responses issued from a $\mathcal{NB}$ distribution with excess of zeros. Write the log-likelihood function of that model.

By defining $\pi_i = \pi(\mathbf{x}_i; \boldsymbol{\beta})$ the probability of “crossing the hurdle” i.e. obtaining an observation different from zero, and $\mu_i = \mu(\mathbf{z}_i; \boldsymbol{\gamma})$ the conditional expectation of the count process, we can provide the expression for the likelihood contributions using the p.m.f. we’ve found on the previous point.

$$\mathbb{P}(Y_i = y_i | \mathbf{x}_i, \mathbf{z}_i; \boldsymbol{\beta}, \boldsymbol{\gamma}) = \begin{cases} 1 - \pi_i & \text{if : } y_i = 0 \\ \pi_i \frac{\Gamma(y_i + \vartheta)}{\Gamma(\vartheta)\Gamma(y_i + 1)} \frac{\vartheta^\vartheta}{(\mu_i + \vartheta)^\vartheta - \vartheta^\vartheta} \left(\frac{\mu_i}{\mu_i + \vartheta} \right)^{y_i} & \text{if : } y_i > 0 \end{cases}$$

Yielding the likelihood :

$$\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\gamma}; \mathbf{y}, \mathbf{X}, \mathbf{Z}) = \prod_{i=1}^n [1 - \pi_i]^{\mathbb{1}(y_i=0)} \left[\pi_i \frac{\Gamma(y_i + \vartheta)}{\Gamma(\vartheta)\Gamma(y_i + 1)} \frac{\vartheta^\vartheta}{(\mu_i + \vartheta)^\vartheta - \vartheta^\vartheta} \left(\frac{\mu_i}{\mu_i + \vartheta} \right)^{y_i} \right]^{\mathbb{1}(y_i>0)}$$

And the log-likelihood :

$$\begin{aligned} \ell(\boldsymbol{\beta}, \boldsymbol{\gamma}; \mathbf{y}, \mathbf{X}, \mathbf{Z}) &= \sum_{i=1}^n \mathbb{1}(y_i = 0) \log(1 - \pi_i) + \mathbb{1}(y_i > 0) \log \pi_i \\ &\quad + \mathbb{1}(y_i > 0) \log \left[\frac{\Gamma(y_i + \vartheta)}{\Gamma(\vartheta)\Gamma(y_i + 1)} \frac{\vartheta^\vartheta}{(\mu_i + \vartheta)^\vartheta - \vartheta^\vartheta} \left(\frac{\mu_i}{\mu_i + \vartheta} \right)^{y_i} \right] \end{aligned}$$

- c) Show that, in a similar fashion as for the Poisson hurdle model, the log-likelihood function computed in **b)** can be separated into two components: the first one for the peak at 0 and the second one for the strictly positive outcomes.

It is clear that we can separate terms according to their dependence on $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ in the last expression, yielding:

$$\begin{aligned} \ell(\boldsymbol{\beta}, \boldsymbol{\gamma}; \mathbf{y}, \mathbf{X}, \mathbf{Z}) &= \sum_{i=1}^n \mathbb{1}(y_i = 0) \log(1 - \pi_i) + \mathbb{1}(y_i > 0) \log \pi_i \\ &\quad + \sum_{i=1}^n \mathbb{1}(y_i > 0) \log \left[\frac{\Gamma(y_i + \vartheta)}{\Gamma(\vartheta)\Gamma(y_i + 1)} \frac{\vartheta^\vartheta}{(\mu_i + \vartheta)^\vartheta - \vartheta^\vartheta} \left(\frac{\mu_i}{\mu_i + \vartheta} \right)^{y_i} \right] \\ \ell(\boldsymbol{\beta}, \boldsymbol{\gamma}; \mathbf{y}, \mathbf{X}, \mathbf{Z}) &= \ell_1(\boldsymbol{\beta}; \mathbf{y}, \mathbf{X}) + \ell_2(\boldsymbol{\gamma}, \vartheta; \mathbf{y}, \mathbf{Z}) \end{aligned}$$

- (a) What does it imply for the estimation process?

The estimation of parameters β and γ can be performed separately using either of these log-likelihood components. It is customary to use a logit link to model the probability of observing a positive count and a log-link to model the count process.

- (b) What does it imply from the point of view of statistical inference?

Both sets of estimates are uncorrelated i.e. $\text{Cov}(\hat{\beta}, \hat{\gamma}) = 0$

2 Exercise 2

Assume that $Y_1, \dots, Y_n$ are responses for n individuals issued from a zero-inflated Poisson model (ZIP) with covariates $x_1, \dots, x_n$ and parameter α explaining zero responses; and covariates $z_1, \dots, z_n$ and parameter δ explaining positive responses.

Let W_i be an unobservable binary variable that indicates the risk state of individual i : $W_i = 1$ if the subject is not at risk of an event; and $W_i = 0$ if the subject is at risk. Notice that Y_i denote the event count for a subject and this event occurs only when $W_i = 0$. We also suppose that $\Pr\{W_i = 1\} = \pi_i$ (where π_i is a function of $x_i^T \alpha$) and $P\{Y_i = y | W_i = 0\} = \lambda_i^y e^{-\lambda_i} / y!$ (where λ_i is a function of $z_i^T \delta$) i.e. $Y_i | W_i = 0$ is issued from the Poisson distribution.

- a) Write the likelihood function

$$L(\alpha, \delta | y_1, \dots, y_n, w_1, \dots, w_n) = \prod_{i=1}^n P\{Y_i = y_i | W_i = w_i\}.$$

Show that the corresponding log-likelihood function $\ell(\alpha, \delta | y_1, \dots, y_n, w_1, \dots, w_n)$ can be written as

$$\ell(\alpha | y_1, \dots, y_n, w_1, \dots, w_n) + \ell(\delta | y_1, \dots, y_n, w_1, \dots, w_n)$$

i.e. the parameters explaining the presence of zeros can be separated from the other parameters. What does it imply numerically?

$$\begin{aligned} L(\alpha, \delta | y_1, \dots, y_n, w_1, \dots, w_n) &= \prod_{i=1}^n P(Y_i = y_i | W_i = w_i) \\ &= \prod_{i=1}^n \pi(x_i^T \alpha)^{w_i} (1 - \pi(x_i^T \alpha))^{1-w_i} \left(\frac{\exp(-\lambda(z_i^T \delta)) \lambda(z_i^T \delta)^{y_i}}{y_i!} \right)^{1-w_i} \end{aligned}$$

Then

$$\begin{aligned} l(\alpha, \delta | y_1, \dots, y_n, w_1, \dots, w_n) &= \sum_{i=1}^n \{w_i \log \pi(x_i^T \alpha) + (1 - w_i) \log(1 - \pi(x_i^T \alpha))\} + \\ &\quad \sum_{i=1}^n \{-(1 - w_i) \lambda(z_i^T \delta) + (1 - w_i) y_i \log \lambda(z_i^T \delta)\} - \\ &\quad \sum_{i=1}^n (1 - w_i) \log y_i! \\ &= l(\alpha | y_1, \dots, y_n, w_1, \dots, w_n) + l(\delta | y_1, \dots, y_n, w_1, \dots, w_n) - \\ &\quad \sum_{i=1}^n (1 - w_i) \log y_i! \end{aligned}$$

In practice the values $w_1, \dots, w_n$ are never observed and the estimators $\hat{\alpha}, \hat{\delta}$ are obtained with the EM-algorithm¹.

With the EM algorithm $W_1, \dots, W_n$ are considered to be missing and are estimated by their expected values $w_1^{(k)}, \dots, w_n^{(k)}$ under the current estimates $\alpha^{(k)}$ and $\delta^{(k)}$ (E-step). Then $\ell(\alpha, \delta | y_1, \dots, y_n, w_1^{(k)}, \dots, w_n^{(k)})$ is maximized (M-step), and the E- and M-steps are iterated until convergence.

More precisely, assuming $\alpha^{(k)}, \delta^{(k)}, w_1^{(k)}, \dots, w_n^{(k)}$ to be the values obtained at the (k) -th iteration, the iteration $(k+1)$ of the EM-algorithm requires the following two steps.

- *E-step.* Find $w_i^{(k+1)}$ the conditional expectation of W_i given $y_i, \alpha^{(k)}, \delta^{(k)}$ i.e. the expectation of W_i with respect to the following probability mass function $P\{W_i | Y_i = y_i, \alpha^{(k)}, \delta^{(k)}\}$.
- *M-step.* Find $\alpha^{(k+1)}, \delta^{(k+1)}$ as $\arg\max_{\alpha, \delta} \ell(\alpha, \delta | y_1, \dots, y_n, w_1^{(k)}, \dots, w_n^{(k)})$.

¹Dempster, A.P., Laird, N.M., and Rubin, D.B. (1977) Maximum Likelihood Estimation from Incomplete Data via the EM Algorithm, JRSS-B

We would like you to provide the detailed description of the E-step, by answering the following questions.

- b) Find the probability $P\{W_i|Y_i = y_i, \alpha^{(k)}, \delta^{(k)}\}$, for short you can write it as $P\{W_i|y_i, \alpha, \delta\}$. In other words find $P\{W_i = 1|Y_i = 0, \alpha^{(k)}, \delta^{(k)}\}$, $P\{W_i = 0|Y_i = 0, \alpha^{(k)}, \delta^{(k)}\}$ and $P\{W_i = 0|Y_i = y_i > 0, \alpha^{(k)}, \delta^{(k)}\}$.

Hint: use the Bayes formula $P\{H|D\} = \frac{P\{D|H\}P\{H\}}{P\{D\}}$.

$$\begin{aligned} P(W_i|y_i, \alpha, \delta) &= \frac{P(Y_i|w_i, \alpha, \delta)P(W_i|\alpha, \delta)}{P(Y_i = y_i|\alpha, \delta)} \\ &= \frac{((\exp\{-\lambda_i\} \lambda_i^{y_i}/y_i!)^{1-w_i} - \mathbb{1}(y_i > 0) \mathbb{1}(w_i = 1)) \pi_i^{w_i} (1 - \pi_i)^{1-w_i}}{(\pi_i + (1 - \pi_i) \exp(-\lambda_i))^{\mathbb{1}(y_i=0)} ((1 - \pi_i) \exp(-\lambda_i) \lambda_i^{y_i}/y_i!)^{\mathbb{1}(y_i>0)}} \\ &= \begin{cases} \frac{\pi_i^{w_i} (1-\pi_i)^{1-w_i} \exp(-\lambda_i(1-w_i))}{\pi_i + (1-\pi_i) \exp(-\lambda_i)} & \text{if } y_i = 0; \\ \frac{\pi_i^{w_i} (1-\pi_i)^{1-w_i} ((\exp(-\lambda_i) \lambda_i^{y_i}/y_i!)^{1-w_i} - \mathbb{1}(w_i=1))}{(1-\pi_i) \exp(-\lambda_i) \lambda_i^{y_i}/y_i!} & \text{if } y_i > 0. \end{cases} \end{aligned}$$

In other words

$$W_i = \begin{cases} 1 & \text{with probability } \frac{\pi_i}{\pi_i + (1-\pi_i) \exp(-\lambda_i)} \text{ if } y_i = 0 \\ 0 & \text{with probability } \frac{(1-\pi_i) \exp(-\lambda_i)}{\pi_i + (1-\pi_i) \exp(-\lambda_i)} \text{ if } y_i = 0 \\ 0 & \text{with probability } 1 \text{ if } y_i > 0 \end{cases}$$

- c) Using $P\{W_i|Y_i, \alpha^{(k)}, \delta^{(k)}\}$ find $w_i^{(k+1)} = E(W_i|Y_i = y_i, \alpha^{(k)}, \delta^{(k)})$.

From the previous point,

$$\begin{aligned} w_i^{k+1} &= E(W_i|Y_i = y_i, \alpha^{(k)}, \delta^{(k)}) \\ &= \frac{\pi_i^{(k)}}{\pi_i^{(k)} + (1 - \pi_i^{(k)}) \exp(-\lambda_i^{(k)})} \end{aligned}$$

- d) How can we interpret the values of $w_1^{(k+1)}, \dots, w_n^{(k+1)}$ obtained at the last iteration of the algorithm ?
The values of $w_i^{(k+1)}, \dots, w_n^{(k+1)}$ can show the probability of being from the spike of a zero.

3 Exercise 3

See the code for the solution.

- a) Source the `truncpoisson.R` code available from the course website to create the `truncpoisson` GLM family object. Fit manually the two steps of the Poisson hurdle model to the data by using the `family=truncpoisson` argument in the `glm` command. Compare your results to the output from the `hurdle` command of the `pscl` package.
- b) Fit a simple Poisson GLM and a ZIP model. Among the three models, which provides the best fit?
- c) Do high insurance levels make the number of consultations increase?